

Within the flat 6D Conformal Spacetime Algebra system ($Cl(4, 1, 1)$), the quantum tunneling effect in a degenerate semiconductor junction—specifically a tunnel diode—can be modeled from a local realist perspective.

Rather than relying on abstract, non-local probability wave function penetration, this model utilizes the **overlapping field envelopes** of extended electron solitons [4.1] and soft-limited potential barriers [Section 3].

Section 5.6: The Tunnel Diode (Esaki Soliton Junction)

A tunnel diode (Esaki diode) is a heavily doped p-n junction where the transition (depletion) layer is extremely narrow. It exhibits a Negative Differential Resistance (NDR) region under forward bias, which this system models through phase-locked soliton transitions across an overlapping barrier.

5.6.1 Structural Parameters of the Junction Due to degenerate doping, the depletion layer width d is thin:

$$d \sim 5 \text{ to } 10 \text{ nm}$$

This physical distance is comparable to the electron soliton’s outer directrix radius r_d [4.1] and the collective softening parameter of the lattice potentials [Section 3]:

$$\delta_{\text{junction}} = r_{f,\text{electron}} + r_{f,\text{lattice}}$$

5.6.2 The Local Realist Tunneling Mechanism In this system, the electron is not a zero-dimensional point charge, but a hollow current loop of radius r_e with an extended bivector field envelope F_{total} [4.1]. The depletion potential barrier is represented by an electrostatic potential profile $V(x)$.

Because the physical depletion barrier width d is of the same order of magnitude as the electron’s loop radius r_e , the effective interaction energy $U_{\text{eff}}(x)$ of the electron as a function of position x across the junction is computed via a continuous spatial convolution of its charge distribution ρ_e :

$$U_{\text{eff}}(x) = \int \rho_e(x' - x) V(x') d^3 x'$$

This convolution smooths out the electrostatic potential barrier. Even when the center of the electron loop is physically on the n-side of the junction, its outer field envelope F_{total} overlaps with the vacant positive potential wells (holes [5.5]) on the p-side. This direct field overlap allows the electron loop to experience an attractive electro-gravitational gradient across the barrier, enabling it to transition to the opposite side without needing to climb over the classical potential peak.

5.6.3 Phase-Locked Transition Current For a transition to occur across the depletion layer, the internal rotation frequency of the electron loop in the n-side conduction state (ω_n) must align with the target valence state frequency on the p-side (ω_p). Under an applied bias voltage V_{bias} , the energy states shift:

$$\hbar\omega_n = \hbar\omega_p + eV_{\text{bias}}$$

The transition current I_{tunnel} is proportional to the spatial overlap of the soliton field envelopes and the density of available empty valence states N_v that are phase-matched with the filled conduction states N_c :

$$I_{\text{tunnel}}(V_{\text{bias}}) = \Gamma_{\text{trans}} \int N_c(E) N_v(E - eV_{\text{bias}}) [f_n(E) - f_p(E - eV_{\text{bias}})] dE$$

where: * Γ_{trans} is the transition rate, which is a function of the spatial convolution of the bivector field envelopes across the depletion width d . * $f_n(E)$ and $f_p(E)$ are the spatial occupancy distributions of the solitons.

5.6.4 Negative Differential Resistance (NDR) As the forward bias voltage V_{bias} is varied, the alignment of the bands changes, giving rise to three distinct regions:

1. **Peak Tunnel Current (I_p at V_p):** As a small forward bias is applied, the occupied conduction states on the n-side align with the unoccupied valence states on the p-side. The phase-matching condition ($\omega_n \approx \omega_p$) is satisfied across a wide energy band, maximizing Γ_{trans} and resulting in a rapid rise in current up to a peak value I_p .
2. **The Negative Resistance Region ($V_p < V_{\text{bias}} < V_v$):** As V_{bias} increases past V_p , the occupied conduction states of the n-side are pushed upward, aligning with the forbidden band gap of the p-side (where there are no localized positive potential wells [5.5]). Because there are no empty states available for phase-locking, the transition rate drops ($\Gamma_{\text{trans}} \rightarrow 0$). The tunnel current decreases as the bias increases, producing the Negative Differential Resistance region:

$$\frac{dI_{\text{tunnel}}}{dV_{\text{bias}}} < 0 \quad \text{for } V_p < V_{\text{bias}} < V_v$$

3. **Classical Diffusion Region ($V_{\text{bias}} > V_v$):** As the bias exceeds the valley voltage V_v , standard thermal diffusion current dominates. The electron solitons gain sufficient kinetic energy to overcome the physical potential barrier through classical convective transport, yielding an exponential current rise:

$$I_{\text{diff}}(V_{\text{bias}}) = I_0 \left(e^{\frac{eV_{\text{bias}}}{k_B T}} - 1 \right)$$

This combined formulation models the tunnel diode's current-voltage characteristic curve purely through the mechanics of overlapping soliton fields and phase-locked transition rates.